

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

2 CORE 200 Reading | Module 2 | Causation Among Variables

2.1 Causal Relativity

2.1.1 Module 2 | Causal Relativity (1 of 2)

Reflect and discuss (A) Causal Relativity: Background Conditions

Consider the claim: “Drinking at least 3 glasses of tap water a day will improve your health.”  1.

- A. Whether your community has enough water
- B. Whether your community’s water supply contains toxic chemicals.
- C. Whether your community is religious or not.
- D. None of the above.

2.1.2 Module 2 | Causal Relativity (2 of 2)

Reflect and discuss (B) Causal Relativity: The Set of Variables

 2.

- A. Exposure to Chicken Pox [yes, no], High Fever [yes, no], Chicken Pox Rash [yes,no].
- B. Exposure to Chicken Pox [yes, no], Infection with Chicken Pox [yes, no], Chicken Pox Rash [yes,no].
- C. Exposure to Chicken Pox [yes, no], Chicken Pox Rash [yes,no].
- D. None of the above.

 3.

- A. All of the potential causes we know about.
- B. All of the potential causes.
- C. All of the actual causes.
- D. None of the above.

2.2 Module 2 | Causal Assignments

Reflect and discuss (C) Causal Assignments

 4.

- A. The values of each of the variables in the causal system besides the effect are observed
- B. The values of each of the variables in the causal system besides the effect are assigned by an intervention
- C. Every conceivable cause of the effect is included
- D. The values of each of the variables in the causal system, including the effect, are assigned by an intervention

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Suppose that our causal system consists of the variables Gas Tank Full [yes, no], Key Turned [yes, no], and Engine Starts [yes, no], where Engine Starts is the effect. ✎ 5.

- A. Cutting the cables from the battery to the spark plugs.
- B. Putting a charged battery and a full tank of gas in the car.
- C. Turning the key.
- D. Emptying the gas tank and turning the key.

2.2.1 Module 2 | Causal Assignments: Intervention vs. Observation

Causal Assignments are Interventions, not Observations

Reflect and discuss (D) Causation in Populations 2200-1

✎ 6.

- A. The background conditions.
- B. A variable designated as the cause
- C. An assignment of a value to each variable in the system besides the effect
- D. All of the above
- E. None of the above

Assume the causal system: {Bitten by Mosquito, Had Inoculation, Has Sickle Cell Gene, Drinks, Gets malaria}. ✎ 7.

- A. Sam was bitten by a mosquito, Sam did not have a malaria inoculation, Sam lives in America.
- B. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene.
- C. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam gets Malaria.
- D. Sam was bitten by a mosquito, Sam did have a malaria inoculation, Sam has the sickle cell gene, and Sam drinks Gin and Tonics.

Consider this set of variables (and values): Smokes [Yes, No]; Nicotine-Stained Fingers [Yes, No]; and Lung Cancer [Yes, No]. ✎ 8.

- A. Smokes = yes, Lung Cancer = No
- B. Smokes = yes, Nicotine-Stained Fingers = Yes, Lung Cancer = No
- C. Smokes = yes
- D. Nicotine-Stained Fingers = No, Lung Cancer = No
- E. Smokes = yes, Nicotine-Stained Fingers = No

2.2.2 Module 2 | The Number of Causal Assignments

Reflect and discuss (E) The Number of Causal Assignments

Consider a causal system with two variables:

- TV [watch none, watch some, watch a lot]
- Fight [never, some, a lot]

 9.

- A. 2
- B. 3
- C. 4
- D. 9

Consider a causal system with three variable

- Meals eaten per day [1,2,3]
- Workouts per day [1,2]
- Weight [Loss, Gain]

Consider the response structure in which the variable Weight is the effect.  10.

- A. 2
- B. 3
- C. 4
- D. 6

2.3 Module 2 | Response Structures

2.3.1 Module 2 | Building Response Structures

2.4 Defining Direct Causation Among Variables

2.4.1 Module 2 | Test Pairs

Reflect and discuss (F) Test Pairs 1

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

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 11.

- A. 1 and 2.
- B. 1 and 3.
- C. 1 and 4.
- D. 2 and 3.
- E. 2 and 4.
- F. 3 and 4.

 Reflect and discuss (G) Test Pairs 2

Consider this response structure:

Causal Assignment	Physical Therapy	Medication	Recovers
1	Yes	Yes	Yes
2	Yes	No	No
3	No	Yes	No
4	No	No	No

 12.

- A. 3 and 4
- B. 1 and 3
- C. 2 and 4
- D. 2 and 3

Consider the following response structure for Passed Exam:

Causal Assignment	Slept 8 Hours	Studied	Passed Exam
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

 13.

- A. 1 and 2
- B. 2 and 3
- C. 1 and 3
- D. 3 and 4

2.4.2 Module 2 | Direct Causation Among Variables

Reflect and discuss (H) Defining Direct Causation Among Variables 1

Assignment	Switch 1	Switch 2	Effect: Garage Light
1	down	down	off
2	down	up	on
3	up	down	off
4	up	up	on

 14.

- A. 1 and 2.
- B. 1 and 3.
- C. 1 and 4.
- D. 2 and 3.
- E. 2 and 4.
- F. 3 and 4.

 15.

- A. Causal assignments 1 and 3.
- B. Causal assignments 2 and 4.
- C. None of the above

Reflect and discuss (I) Variable Causation Quiz

 16.

- A. Y differs in every test pair for X
- B. There is at least one test pair for X in the response structure for Y across which Y differs
- C. There is at least one test pair for X in the response structure for Y across which Y does not differ
- D. There is at least one test pair for X in the response structure for Y

Consider the following response structure:

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

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17.

- A. 2 and 3 is a test pair for Starter Broken in which Car Starts differs
- B. 3 and 4 is a test pair for Starter Broken in which Car Starts differs
- C. 1 and 3 is a test pair for Starter Broken in which Car Starts differs
- D. 2 and 4 is a test pair for Starter Broken in which Car Starts differs

Consider the following response structure:

Causal Assignment	X	Y	Z
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	No
4	No	No	No

18.

- A. X only
- B. Y only
- C. Both X and Y
- D. Neither X nor Y

2.4.3 Module 2 | Subtleties

Reflect and discuss (J) Causation: Subtleties 1

Response Structure for the Light Bulb

Assignment	Switch	Battery	Effect: Light Bulb
1	down	charged	Off
2	down	uncharged	Off
3	up	charged	On
4	up	uncharged	Off

19.

- A. 1 and 2.
- B. 1 and 3.
- C. 1 and 4.
- D. 2 and 3.
- E. 2 and 4.
- F. 3 and 4.

Reflect and discuss (K) Causation: Subtleties 2

20.

- A. 1 and 2.
- B. 1 and 3.
- C. 1 and 4.
- D. 9 and 10.
- E. 15 and 16.
- F. 1 and 9.

All the following are test pairs of causal assignments for the variable Drinker.  21.

- A. 1 and 2
- B. 3 and 4
- C. 5 and 6
- D. 7 and 8
- E. 9 and 10
- F. 11 and 12
- G. 13 and 14
- H. 15 and 16
- I. None of the above.

 22.

- A. Yes
- B. No
- C. Impossible to tell.

2.4.4 Module 2 | Interactions

 Reflect and discuss (L) Interactions 1

 23.

- A. Yes
- B. No
- C. Not enough information to tell

Consider the following response structure:

Causal Assignment	Starter Broken	Battery Charged	Car Starts
1	Yes	Yes	No
2	Yes	No	No
3	No	Yes	Yes
4	No	No	No

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 24.

- A. Both are causes of Car Starts
- B. The influence of Battery Charged on the Car Starts depends on the value of Starter Broken
- C. Starter Broken is not a cause of Car Starts
- D. They have the same test pairs

2.5 Causation in Populations

2.5.1 Module 2 | Individuals and Populations

2.5.2 Module 2 | Response Structure Uniformity

Reflect and discuss (M) Response Structure Uniformity

Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to whether their gas tanks explode in rear-end collisions or not (because of a structural defect in the way that 1993 Escorts were built)? (choose one)

- A. Yes
- B. No

Is it likely that the population of all 1993 Ford Escorts exhibit response structure uniformity with respect to exposure to the sun causing their upholstery to crack? (choose one)

- A. Yes
- B. No

Reflect and discuss (N) Link it to your research proposal

Many of the ideas you have learned in this handout will be relevant to your Semester Research Project. Discuss the connection of these ideas to your project here in as much detail as possible, seeking help from your instructor when needed. (Refer back to Barroga and Matanguihan (2022) for more help. If your project has not progressed to this stage, you should come back to this activity later.)
